



Project: **Lürssen-Kröger Werft GmbH**
Schacht-Audorf, Germany
Hull No. 13759, project "Cosmos"

Client: **SCHIFFSDIESELTECHNIK KIEL GMBH**
Kieler Straße 177
24768 Rendsburg

Office: **Kiel, Germany**

Client's Order Number: **BD21/000378**
PR20/09-3

Date: **12 January 2022**

Order Status: **Complete**

Inspection Dates

First: **17 December 2021**

Final: **21 December 2021**

This certificate is issued to

The above client to certify that at their request the undersigned surveyor to Lloyd's Register EMEA did attend their works on the date(s) shown for the purpose of examining the item(s) mentioned below:

ONE (1) AUXILIARY MARINE DIESEL GENERATOR MY 'Cosmos' – DG3
INCLUDING MTU DIESEL ENGINE TYPE 16V2000 M41B

Manufacturer: SDT Schiffsdieseltechnik GmbH
Kieler Straße 177
24768 Rendsburg
Germany

Consisting of:

1 – Diesel Engine

Type: MTU 16V2000 M41B
Make: MTU Friedrichshafen GmbH, 88045 Friedrichshafen, Germany
Description: 930 kW at 1800 rpm
Serial No.: 536 113 910
Covered with LR QAM 030 certificate HAM2152312 dated 29 September 2021 attached hereto.

1 - Generator

Type: LSAM 50.2 L8 C 29S/4
Make: M.L.S. Holice spol. s r.o, Olomouc, Czech Republic
Description: 1090 kVA, 872 kW, 690 V, 60 Hz, 1800 rpm
Serial No.: 38231000002
Covered with LR certificate PRA2100408/2/A1 dated 07 September 2021 attached hereto.

1 - Coupling

Type: Vulkan VULASTIK-L 3413
Make: Vulkan Kupplungs- und Getriebebau B. Hackforth GmbH, Herne, Germany
Description: 930 kW, 1800 rpm
Serial No.: 312100524-3
Covered with LR certificate HAM2152875/1 dated 01 September 2021 attached hereto.

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1 - Local Operation Panel (LOP) / Engine Control System

Make:	MTU Friedrichshafen GmbH, Germany
Type:	LOP 10-03
Panel No.:	264535038
Control Unit Type:	SAM1-07
Serial No.:	264534975
EMU No.:	263533419
ECU No.:	263534044

The generator unit has been examined on completion of alignment and assembly, fitted on a common foundation frame.
The workmanship appeared to be satisfactory.

The generator unit was subject of factory acceptance tests, which have been carried out with satisfactory results.

For results of load tests, electric load steps and check of alarm and safety systems see attached FAT protocols dated 17.12.2021.

The Torsional Vibration Calculation has been approved with document of Hamburg Technical Support Office:
HTS/ENS 43813-21, issue number 0 dated 21 September 2021.

The Resilient Mounted Generator has been approved with document of Technical Support Office, Rotterdam:
RTS-ENG-214215, issue number 0 dated 03 December 2021.

The Monitoring, Control and Safety System has been approved with document of Hamburg Technical Support Office:
HTS/ETS 42147-22, issue number 0 dated 20 January 2021.

For further identification purposes the unit has been stamped on the foundation frame:

LR HAM 2040110
12 HP 21

